# Experiment-10 Write the python program to implement A* algorithm

EXP-10 AI.py - C:/Users/prana/OneDrive/Desktop/EXP-10 AI.py (3.13.14)

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```
graph = {
    'A': [('B', 1), ('C', 3)],
    'B': [('D', 2)],
    'C': [('D', 1)],
    'D': []
}

h = {'A': 4, 'B': 2, 'C': 1, 'D': 0}

def astar(start, goal):
    open_list = [(start, 0, [start])]

    while open_list:
        node, cost, path = open_list.pop(0)

        if node == goal:
            print("Path:", path)
            print("Cost:", cost)
            return

        for neighbor, distance in graph[node]:
            new_cost = cost + distance
            open_list.append((neighbor, new_cost, path + [neighbor]))

        open_list.sort(key=lambda x: x[1] + h[x[0]])

astar('A', 'D')
```

IDLE Shell 3.13.14

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```
Python 3.13.14 (tags/v3.13.14:fd17997, Jun 10 2026, 13:03:48) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
========== RESTART: C:/Users/prana/OneDrive/Desktop/EXP-10 AI.py ==========
Path: ['A', 'B', 'D']
Cost: 3
>>>
```